

SANGOLA MAHAVIDYALAYA, SANGOLA

Department of Computer Science and Application

Subject: .NET Core | Practical Assignment - 01 | Date: 29/07/2025

Q1. Write a C# Program to perform basic arithmetic operations on two numbers.

```
using System;

class Q1_ArithmeticOps {
    static void Main() {
        int a = 10, b = 5;
        Console.WriteLine("Sum = " + (a + b));
        Console.WriteLine("Diff = " + (a - b));
        Console.WriteLine("Product = " + (a * b));
        Console.WriteLine("Division = " + (a / b));
        Console.WriteLine("Mod = " + (a % b));
    }
}

// Output: Sum=15, Diff=5, Product=50, Division=2, Mod=0
```

Q2. Write a C# Program to find average of three numbers.

```
using System;

class Q2_Average {
    static void Main() {
        double a = 10, b = 20, c = 30;
        double avg = (a + b + c) / 3;
        Console.WriteLine("Average = " + avg);
    }
}

// Output: Average = 20
```

Q3. Write a C# Program to calculate perimeter (circumference) of a circle.

```
using System;

class Q3_Circumference {
    static void Main() {
        double radius = 7;
        double circumference = 2 * Math.PI * radius;
        Console.WriteLine("Circumference = " + Math.Round(circumference, 2));
    }
}

// Output: Circumference = 43.96
```

Q4. Write a C# Program to calculate area of a circle.

```
using System;

class Q4_CircleArea {
    static void Main() {
        double radius = 7;
        double area = Math.PI * radius * radius;
        Console.WriteLine("Area = " + Math.Round(area, 2));
    }
}
```

```
}  
}  
// Output: Area = 153.94
```

Q5. Write a C# Program to calculate area of a triangle.

```
using System;  
class Q5_TriangleArea {  
static void Main() {  
double b = 10, h = 5;  
double area = 0.5 * b * h;  
Console.WriteLine("Area = " + area);  
}  
}  
// Output: Area = 25
```

Q6. Write a C# Program to calculate area of a square.

```
using System;  
class Q6_SquareArea {  
static void Main() {  
double side = 4;  
double area = side * side;  
Console.WriteLine("Area = " + area);  
}  
}  
// Output: Area = 16
```

Q7. Write a C# Program to find max number between two numbers using ternary operator.

```
using System;  
class Q7_MaxTernary {  
static void Main() {  
int a = 5, b = 10;  
int max = (a > b) ? a : b;  
Console.WriteLine("Max = " + max);  
}  
}  
// Output: Max = 10
```

Q8. Check positive/negative, max of two, and leap year.

```
using System;  
class Q8_MultiCheck {  
static void Main() {  
// Positive/Negative  
int n = -5;  
Console.WriteLine(n > 0 ? "Positive" : "Negative");  
  
// Max of two  
int a = 10, b = 20;  
Console.WriteLine("Max = " + (a > b ? a : b));  
}
```

```
// Leap Year
int year = 2024;
bool isLeap = (year % 4 == 0 && year % 100 != 0) || (year % 400 == 0);
Console.WriteLine(year + " is " + (isLeap ? "a Leap Year" : "Not a Leap Year"));
}
}
// Output: Negative, Max=20, 2024 is a Leap Year
```

Q9. Menu-driven program for arithmetic operations using switch statement.

```
using System;
class Q9_Switch {
static void Main() {
int a = 10, b = 5;
char choice = '+';
switch (choice) {
case '+': Console.WriteLine("Sum = " + (a + b)); break;
case '-': Console.WriteLine("Diff = " + (a - b)); break;
case '*': Console.WriteLine("Product = " + (a * b)); break;
case '/': Console.WriteLine("Division = " + (a / b)); break;
case '%': Console.WriteLine("Mod = " + (a % b)); break;
default: Console.WriteLine("Invalid choice"); break;
}
}
}
// Output: Sum = 15
```

Q10. Area of circle, square, rectangle using switch statement.

```
using System;
class Q10_AreaSwitch {
static void Main() {
int shape = 1; // 1=Circle, 2=Square, 3=Rectangle
switch (shape) {
case 1:
double r = 7;
Console.WriteLine("Circle Area = " + Math.Round(Math.PI * r * r, 2));
break;
case 2:
double s = 4;
Console.WriteLine("Square Area = " + (s * s));
break;
case 3:
double l = 5, b = 6;
Console.WriteLine("Rectangle Area = " + (l * b));
break;
}
}
}
// Output: Circle Area = 153.94
```

Q11. Generate electricity bill using else-if ladder.

```
using System;
class Q11_ElectricityBill {
static void Main() {
int units = 150;
double bill = 0;
if (units <= 80)
bill = units * 5;
else if (units <= 150)
bill = (80 * 5) + ((units - 80) * 7);
else
bill = (80 * 5) + (70 * 7) + ((units - 150) * 10);
Console.WriteLine("Electricity Bill = Rs." + bill);
}
}
// Output for 150 units: Rs.975
```

Q12. Calculate library fine using else-if ladder.

```
using System;
class Q12_LibraryFine {
static void Main() {
int daysLate = 7;
double fine = 0;
if (daysLate <= 5)
fine = 0;
else if (daysLate <= 10)
fine = daysLate * 5;
else
fine = daysLate * 10;
Console.WriteLine("Fine Amount = Rs." + fine);
}
}
// Output for 7 days: Rs.35
```

Q13. Find minimum number between two numbers.

```
using System;
class Q13_MinTwo {
static void Main() {
int a = 10, b = 5;
if (a == b)
Console.WriteLine("Both Equal");
else
Console.WriteLine("Min = " + (a < b ? a : b));
}
}
// Output: Min = 5
```

Q14. Find maximum number between two numbers.

```
using System;
```

```
class Q14_MaxTwo {
static void Main() {
int a = 15, b = 25;
Console.WriteLine("Max = " + (a > b ? a : b));
}
}
// Output: Max = 25
```

Q15. Find minimum number between three numbers.

```
using System;
class Q15_MinThree {
static void Main() {
int a = 4, b = 8, c = 6;
int min = a;
if (b < min) min = b;
if (c < min) min = c;
Console.WriteLine("Min = " + min);
}
}
// Output: Min = 4
```

Q16. Find maximum number between three numbers.

```
using System;
class Q16_MaxThree {
static void Main() {
int a = 3, b = 5, c = 2;
int max = a;
if (b > max) max = b;
if (c > max) max = c;
Console.WriteLine("Max = " + max);
}
}
// Output: Max = 5
```

Q17. Check if entered number is even or odd.

```
using System;
class Q17_EvenOdd {
static void Main() {
int n = 4;
Console.WriteLine(n % 2 == 0 ? n + " is Even" : n + " is Odd");
}
}
// Output: 4 is Even
```

Q18. Check if number is even or odd using conditional operator.

```
using System;
class Q18_EvenOddTernary {
static void Main() {
int n = 17;
```

```
string result = (n % 2 == 0) ? "Even" : "Odd";  
Console.WriteLine(n + " is " + result);  
}  
}  
// Output: 17 is Odd
```

Q19. Print odd numbers between 1 to 500.

```
using System;
class Q19_OddNumbers {
static void Main() {
Console.Write("Odd Numbers: ");
for (int i = 1; i <= 500; i += 2)
Console.Write(i + " ");
Console.WriteLine();
}
}
// Output: 1 3 5 7 ... 499
```

Q20. Print even numbers between 1 to 500.

```
using System;
class Q20_EvenNumbers {
static void Main() {
Console.Write("Even Numbers: ");
for (int i = 2; i <= 500; i += 2)
Console.Write(i + " ");
Console.WriteLine();
}
}
// Output: 2 4 6 8 ... 500
```

Q21. Add all odd numbers from 1 to 20.

```
using System;
class Q21_SumOdd {
static void Main() {
int sum = 0;
for (int i = 1; i <= 20; i += 2)
sum += i;
Console.WriteLine("Sum of odd numbers = " + sum);
}
}
// Output: Sum of odd numbers = 100
```

Q22. Add all even numbers from 1 to 20.

```
using System;
class Q22_SumEven {
static void Main() {
int sum = 0;
for (int i = 2; i <= 20; i += 2)
sum += i;
Console.WriteLine("Sum of even numbers = " + sum);
}
}
// Output: Sum of even numbers = 110
```

Q23. Calculate grade of student based on obtained marks.

```

using System;
class Q23_Grade {
static void Main() {
int marks = 85;
if (marks >= 80)
Console.WriteLine("Grade A");
else if (marks >= 70)
Console.WriteLine("Grade B");
else if (marks >= 60)
Console.WriteLine("Grade C");
else if (marks >= 45)
Console.WriteLine("Grade D");
else
Console.WriteLine("Fail");
}
}
// Output for 85: Grade A

```

Q24. Demonstrate Type Inference.

```

using System;
class Q24_TypeInference {
static void Main() {
var intVal = 10; // int
var dblVal = 3.14; // double
var strVal = "Hello"; // string
var boolVal = true; // bool
Console.WriteLine("int: " + intVal);
Console.WriteLine("double: " + dblVal);
Console.WriteLine("string: " + strVal);
Console.WriteLine("bool: " + boolVal);
Console.WriteLine("Types: " + intVal.GetType() + ", " +
dblVal.GetType() + ", " + strVal.GetType());
}
}

```

Q25. Demonstrate Console I/O operations in C#.

```

using System;
class Q25_ConsoleIO {
static void Main() {
Console.Write("Enter your name: ");
string name = Console.ReadLine();
Console.Write("Enter your age: ");
int age = int.Parse(Console.ReadLine());
Console.WriteLine("Hello, " + name + "! You are " + age + " years old.");
}
}

```

Q26. Demonstrate printing concatenated string using formatted string.

```

using System;

```



```

class Q26_FormattedString {
static void Main() {
string name = "Alice";
int age = 20;
double marks = 85.5;
// String interpolation
Console.WriteLine($"Name: {name}, Age: {age}, Marks: {marks}");
// String.Format
Console.WriteLine(String.Format("Name: {0}, Age: {1}, Marks: {2}", name, age, marks));
// Concatenation
Console.WriteLine("Name: " + name + ", Age: " + age + ", Marks: " + marks);
}
}

```

Q27. Perform basic Arithmetic operations (WAP).

```

using System;
class Q27_BasicArithmetic {
static void Main() {
Console.Write("Enter first number: ");
double a = double.Parse(Console.ReadLine());
Console.Write("Enter second number: ");
double b = double.Parse(Console.ReadLine());
Console.WriteLine($"Addition = {a + b}");
Console.WriteLine($"Subtraction = {a - b}");
Console.WriteLine($"Multiplication = {a * b}");
Console.WriteLine($"Division = {(b != 0 ? a / b : double.NaN)}");
Console.WriteLine($"Modulus = {a % b}");
}
}

```

Q28. Demonstrate Ternary Operator.

```

using System;
class Q28_TernaryOperator {
static void Main() {
int a = 15, b = 10;
int max = (a > b) ? a : b;
int min = (a < b) ? a : b;
string result = (a % 2 == 0) ? "Even" : "Odd";
Console.WriteLine("Max = " + max);
Console.WriteLine("Min = " + min);
Console.WriteLine(a + " is " + result);
}
}

```

Q29. Demonstrate Type Conversion.

```

using System;
class Q29_TypeConversion {
static void Main() {
// Implicit conversion

```

```
int intVal = 42;
double dblVal = intVal;
Console.WriteLine("Implicit int to double: " + dblVal);

// Explicit (casting)
double pi = 3.14159;
int piInt = (int)pi;
Console.WriteLine("Explicit double to int: " + piInt);

// Convert class
string strNum = "100";
int parsed = Convert.ToInt32(strNum);
Console.WriteLine("String to int: " + parsed);

// int.Parse
int parsed2 = int.Parse("200");
Console.WriteLine("Parse: " + parsed2);

// ToString
int n = 55;
string s = n.ToString();
Console.WriteLine("int to string: " + s);
}
```

Q30. Find factorial of entered number.

```
using System;
class Q30_Factorial {
static void Main() {
int n = 5;
long fact = 1;
for (int i = 1; i <= n; i++)
fact *= i;
Console.WriteLine("Factorial of " + n + " is " + fact);
}
}
// Output: Factorial of 5 is 120
// Note: Factorial of 0 is 1 (by definition)
```

Q31. Find sum of digits of entered number.

```
using System;
class Q31_SumOfDigits {
static void Main() {
int n = 123, sum = 0, temp = Math.Abs(n);
while (temp > 0) {
sum += temp % 10;
temp /= 10;
}
Console.WriteLine("Sum of digits of " + n + " = " + sum);
}
}
// Output: Sum of digits of 123 = 6
```

Q32. Check if entered number is Armstrong or not.

```
using System;
class Q32_Armstrong {
static void Main() {
int n = 153;
int temp = n, sum = 0, digits = n.ToString().Length;
while (temp > 0) {
int d = temp % 10;
sum += (int)Math.Pow(d, digits);
temp /= 10;
}
Console.WriteLine(n + (sum == n ? " is an Armstrong number." : " is not an Armstrong number."));
}
}
// Output: 153 is an Armstrong number.
```

Q33. Find all Armstrong numbers between 1 to 500.

```
using System;
class Q33_AllArmstrong {
static void Main() {
```

```

Console.Write("Armstrong numbers between 1 to 500: ");
for (int n = 1; n <= 500; n++) {
int temp = n, sum = 0, digits = n.ToString().Length;
while (temp > 0) {
int d = temp % 10;
sum += (int)Math.Pow(d, digits);
temp /= 10;
}
if (sum == n) Console.Write(n + " ");
}
Console.WriteLine();
}
}
// Output: 1 2 3 4 5 6 7 8 9 153 370 371 407

```

Q34. Print reverse of entered number.

```

using System;
class Q34_ReverseNumber {
static void Main() {
int n = 1234, rev = 0, temp = n;
while (temp != 0) {
rev = rev * 10 + temp % 10;
temp /= 10;
}
Console.WriteLine("Reversed number is " + rev);
}
}
// Output: Reversed number is 4321

```

Q35. Check if entered number is palindrome or not.

```

using System;
class Q35_Palindrome {
static void Main() {
int n = 121, rev = 0, temp = n;
while (temp != 0) {
rev = rev * 10 + temp % 10;
temp /= 10;
}
Console.WriteLine(n + (rev == n ? " is a palindrome." : " is not a palindrome.));
}
}
// Output: 121 is a palindrome.

```

Q36. Find all palindrome numbers between 1 to 500.

```

using System;
class Q36_AllPalindromes {
static void Main() {
Console.Write("Palindrome numbers between 1 to 500: ");
for (int n = 1; n <= 500; n++) {

```

```

int rev = 0, temp = n;
while (temp != 0) { rev = rev * 10 + temp % 10; temp /= 10; }
if (rev == n) Console.Write(n + " ");
}
Console.WriteLine();
}
}
// Output: 1 2 3 ... 9 11 22 33 ... 99 101 111 ... 494 505 (all palindromes)

```

Q37. Display Fibonacci series up to given number.

```

using System;
class Q37_Fibonacci {
static void Main() {
int limit = 50;
int a = 0, b = 1;
Console.Write("Fibonacci series up to " + limit + ": ");
while (a <= limit) {
Console.Write(a + " ");
int next = a + b;
a = b;
b = next;
}
Console.WriteLine();
}
}
// Output: 0 1 1 2 3 5 8 13 21 34

```

Q38. Calculate gross salary.

```

using System;
class Q38_GrossSalary {
static void Main() {
double basic = 10000;
double da = 0.40 * basic;
double hra = 0.20 * basic;
double ppf = 0.10 * basic;
double gross = basic + da + hra - ppf;
Console.WriteLine("Basic Salary = " + basic);
Console.WriteLine("DA (40%) = " + da);
Console.WriteLine("HRA (20%) = " + hra);
Console.WriteLine("PPF (10%) = " + ppf);
Console.WriteLine("Gross Salary = " + gross);
}
}
// Output: Gross Salary = 15000

```

Q39. Display Floyd's Triangle pattern (1 / 01 / 101 / 0101 / 10101).

```
using System;
class Q39_FloydsTriangle {
static void Main() {
int rows = 5;
for (int i = 1; i <= rows; i++) {
int start = (i % 2 == 0) ? 0 : 1;
for (int j = 1; j <= i; j++) {
Console.Write(start);
start = 1 - start;
}
Console.WriteLine();
}
}
}
/* Output:
1
01
101
0101
10101 */
```

Q40. Display given patterns (multiple patterns).

```
using System;
class Q40_Pattern1 { // Pattern: 0 / 10 / 210 / ... (left triangle descending)
static void Main() {
int rows = 9;
for (int i = 0; i < rows; i++) {
for (int j = i; j >= 0; j--)
Console.Write(j + " ");
Console.WriteLine();
}
}
}
```

```
using System;
class Q40_Pattern2 { // Pattern: 1 / 1 2 / 1 2 3 / ...
static void Main() {
for (int i = 1; i <= 5; i++) {
for (int j = 1; j <= i; j++)
Console.Write(j + " ");
Console.WriteLine();
}
}
}
```

```
using System;
class Q40_Pattern3 { // Pattern: 5 4 3 2 1 / 5 4 3 2 / 5 4 3 / 5 4 / 5
static void Main() {
int n = 5;
```

```

for (int i = n; i >= 1; i--) {
for (int j = n; j >= n - i + 1; j--)
Console.Write(j + " ");
Console.WriteLine();
}
}
}

using System;
class Q40_Pattern4 { // Right triangle: 1 / 2 3 / 4 5 6 / 7 8 9 10
static void Main() {
int num = 1;
for (int i = 1; i <= 4; i++) {
for (int j = 1; j <= i; j++)
Console.Write(num++ + " ");
Console.WriteLine();
}
}
}

using System;
class Q40_Pattern5 { // 1/2 3/1 2 3 repeated pattern
static void Main() {
int rows = 9, num = 1;
for (int i = 1; i <= rows; i++) {
Console.Write("1");
if (i > 1) { Console.Write(" 2"); num = 3; }
if (i > 2) for (int j = 3; j <= i; j++) Console.Write(" " + j);
Console.WriteLine();
}
}
}

using System;
class Q40_Pattern6 { // @ triangle and $ right-angle triangle
static void Main() {
// @ pyramid
int rows = 7;
for (int i = 1; i <= rows; i++) {
for (int j = 1; j <= i; j++) Console.Write("@");
Console.WriteLine();
}
Console.WriteLine();
// $ right-triangle (decreasing)
for (int i = 10; i >= 1; i--) {
for (int j = 1; j <= i; j++) Console.Write("$ ");
Console.WriteLine();
}
}
}

```

Q41. Demonstrate Boxing and UnBoxing.

```
using System;
class Q41_BoxingUnboxing {
static void Main() {
int val = 42;

// Boxing: value type -> object (reference type)
object boxed = val;
Console.WriteLine("Boxed value: " + boxed);
Console.WriteLine("Type after boxing: " + boxed.GetType());

// Unboxing: object -> value type
int unboxed = (int)boxed;
Console.WriteLine("Unboxed value: " + unboxed);

// Modify original - boxed copy not affected
val = 100;
Console.WriteLine("Original after change: " + val);
Console.WriteLine("Boxed still: " + boxed);
}
}
```

Q42. Demonstrate Enumeration.

```
using System;
enum Days { Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, Saturday }
enum Season { Spring = 1, Summer, Autumn, Winter }

class Q42_Enum {
static void Main() {
Days today = Days.Wednesday;
Console.WriteLine("Today: " + today);
Console.WriteLine("Day number: " + (int)today);

Season s = Season.Summer;
Console.WriteLine("Season: " + s + " = " + (int)s);

// Loop through enum
foreach (Days d in Enum.GetValues(typeof(Days)))
Console.WriteLine((int)d + " -> " + d);
}
}
```

Q43. Perform sum of array elements.

```
using System;
class Q43_SumArray {
static void Main() {
int[] arr = { 10, 20, 30, 40, 50 };
int sum = 0;
foreach (int x in arr)
```



```
sum += x;
Console.WriteLine("Array: " + string.Join(", ", arr));
Console.WriteLine("Sum = " + sum);
}
}
// Output: Sum = 150
```

Q44. Demonstrate various operations on Array using in-built methods.

```
using System;
class Q44_ArrayMethods {
static void Main() {
int[] arr = { 5, 3, 8, 1, 9, 2, 7 };
Console.WriteLine("Original: " + string.Join(", ", arr));

Array.Sort(arr);
Console.WriteLine("Sorted: " + string.Join(", ", arr));

Array.Reverse(arr);
Console.WriteLine("Reversed: " + string.Join(", ", arr));

Console.WriteLine("Length: " + arr.Length);
Console.WriteLine("Max: " + arr[0]); // after reverse sort
Console.WriteLine("Min: " + arr[arr.Length - 1]);

int idx = Array.IndexOf(arr, 5);
Console.WriteLine("Index of 5: " + idx);

int[] copy = (int[])arr.Clone();
Console.WriteLine("Clone: " + string.Join(", ", copy));
}
}
```

Q45. Demonstrate creation of Tuple using Parenthesis Method.

```
using System;
class Q45_TupleParenthesis {
static void Main() {
// Tuple using parenthesis (ValueTuple)
var student = (Name: "Alice", Age: 20, Marks: 85.5);
Console.WriteLine("Name: " + student.Name);
Console.WriteLine("Age: " + student.Age);
Console.WriteLine("Marks: " + student.Marks);

var point = (10, 20); // unnamed
Console.WriteLine("Point: (" + point.Item1 + ", " + point.Item2 + ")");
}
}
```

Q46. Demonstrate creation of Tuple using Create() Method.

```
using System;
class Q46_TupleCreate {
static void Main() {
// Tuple using Tuple.Create()
Tuple<string, int, double> t = Tuple.Create("Bob", 21, 78.5);
Console.WriteLine("Name: " + t.Item1);
Console.WriteLine("Age: " + t.Item2);
Console.WriteLine("Marks: " + t.Item3);

var t2 = Tuple.Create(1, "Hello", true, 3.14);
Console.WriteLine(t2.Item1 + " " + t2.Item2 + " " + t2.Item3 + " " + t2.Item4);
}
}
```

Q47. Demonstrate Nested Tuple.

```
using System;
class Q47_NestedTuple {
static void Main() {
// Nested Tuple
var nested = Tuple.Create("Alice", 20,
Tuple.Create("Mumbai", "CS", 9.1));

Console.WriteLine("Name: " + nested.Item1);
Console.WriteLine("Age: " + nested.Item2);
Console.WriteLine("City: " + nested.Item3.Item1);
Console.WriteLine("Branch: " + nested.Item3.Item2);
Console.WriteLine("CGPA: " + nested.Item3.Item3);
}
}
```

Q48. Demonstrate Tuple as Method Parameter.

```
using System;
class Q48_TupleParameter {
```

```

static void DisplayStudent(Tuple<string, int, double> s) {
    Console.WriteLine($"Name: {s.Item1}, Age: {s.Item2}, Marks: {s.Item3}");
}

static double CalcAverage((double a, double b, double c) t) {
    return (t.a + t.b + t.c) / 3.0;
}

static void Main() {
    var student = Tuple.Create("Charlie", 22, 90.5);
    DisplayStudent(student);

    var marks = (85.0, 90.0, 78.0);
    Console.WriteLine("Average: " + CalcAverage(marks));
}

```

Q49. Demonstrate returning Tuple from Method.

```

using System;

class Q49_ReturnTuple {
    static (int min, int max, double avg) GetStats(int[] arr) {
        int min = arr[0], max = arr[0], sum = 0;
        foreach (int x in arr) {
            if (x < min) min = x;
            if (x > max) max = x;
            sum += x;
        }
        return (min, max, (double)sum / arr.Length);
    }

    static void Main() {
        int[] data = { 4, 7, 2, 9, 1, 5 };
        var result = GetStats(data);
        Console.WriteLine("Min: " + result.min);
        Console.WriteLine("Max: " + result.max);
        Console.WriteLine("Avg: " + result.avg);
    }
}

```

Q50. Store even items and odd items in separate arrays from 1D array.

```

using System;
using System.Collections.Generic;

class Q50_EvenOddArrays {
    static void Main() {
        int[] arr = { 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 };
        List<int> evens = new List<int>();
        List<int> odds = new List<int>();

        foreach (int x in arr) {
            if (x % 2 == 0) evens.Add(x);
            else odds.Add(x);
        }
    }
}

```

```

Console.WriteLine("Original: " + string.Join(", ", arr));
Console.WriteLine("Even: " + string.Join(", ", evens));
Console.WriteLine("Odd: " + string.Join(", ", odds));
}
}
// Even: 2 4 6 8 10 Odd: 1 3 5 7 9

```

Q51. Perform Addition of TWO matrices.

```

using System;
class Q51_MatrixAddition {
static void Main() {
int[,] A = { {1, 2, 3}, {4, 5, 6}, {7, 8, 9} };
int[,] B = { {9, 8, 7}, {6, 5, 4}, {3, 2, 1} };
int rows = A.GetLength(0), cols = A.GetLength(1);
int[,] C = new int[rows, cols];

for (int i = 0; i < rows; i++)
for (int j = 0; j < cols; j++)
C[i, j] = A[i, j] + B[i, j];

Console.WriteLine("Matrix A:");
PrintMatrix(A, rows, cols);
Console.WriteLine("Matrix B:");
PrintMatrix(B, rows, cols);
Console.WriteLine("Sum (A + B):");
PrintMatrix(C, rows, cols);
}

static void PrintMatrix(int[,] M, int r, int c) {
for (int i = 0; i < r; i++) {
for (int j = 0; j < c; j++)
Console.Write(M[i, j] + " ");
Console.WriteLine();
}
}
}
// Each row of C = sum of corresponding rows of A and B

```